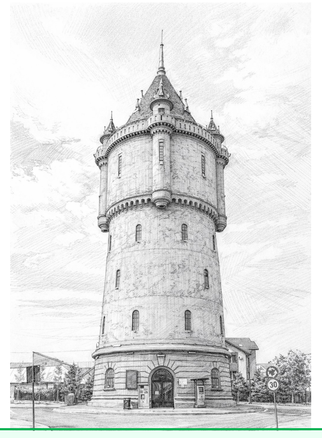
COSIN
SUSTAINABILITY & INNOVATION

Drobeta - Turnu Severin, May 28-30, 2026



ANGIE
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Ultrasonic welding

Ultrasonic welding (USW) is a solid-state joining technology that has attracted significant attention in recent years due to its high energy efficiency, extremely short processing time, and ability to join both similar and dissimilar materials without macroscopic melting.

Compared with conventional fusion-based joining methods, USW requires lower energy input, generates a very limited heat-affected zone, and eliminates the need for filler materials, shielding gases, or chemical fluxes. These characteristics make ultrasonic welding particularly attractive from a sustainability perspective, supporting the development of cleaner and more resource-efficient manufacturing technologies.

The growing interest in ultrasonic welding is especially evident in the electrical and electronic sectors, where the process is increasingly used for conductor joining, electrical interconnections, battery tab-to-busbar welding, and flexible electronic systems. In such applications, USW offers important environmental and technological advantages, including reduced energy consumption, lower thermal degradation of conductive materials, minimal material waste, and improved recyclability of joined components.

Furthermore, the solid-state nature of the process reduces the formation of brittle intermetallic compounds and improves the electrical conductivity and long-term reliability of the connections.

For the welded joints presented above, the heat source can be considered as a combined thermo-mechanical source specific to ultrasonic welding of copper conductors, consisting of frictional heat generation at the interface, heat produced by plastic deformation, and energy dissipation caused by ultrasonic vibrations.

The total heat generation can be expressed as:

$$Q_{tot} = Q_f + Q_p + Q_u$$

frictional heat generated at the interface between the Cu conductors

$$Q_f = \mu \cdot p \cdot v_s = \mu \cdot p \cdot 2\pi \, \mathrm{f} \, \mathrm{A}$$

The heat generated by plastic deformation

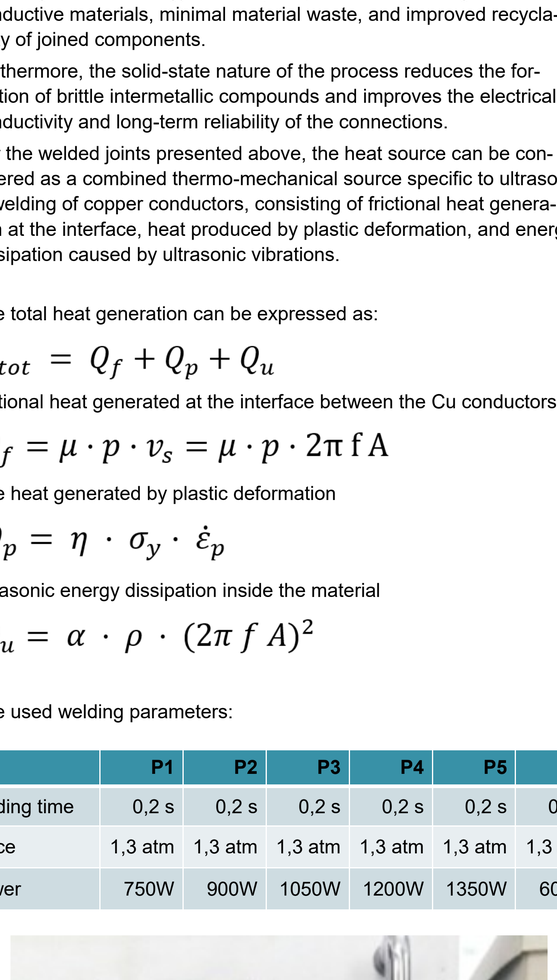
$$Q_p = \eta \cdot \sigma_y \cdot \dot{\varepsilon}_p$$

ultrasonic energy dissipation inside the material

$$Q_u = \alpha \cdot \rho \cdot (2\pi f A)^2$$

The used welding parameters:

	P1	P2	P3	P4	P5	P6
Holding time	0,2 s	0,2 s	0,2 s	0,2 s	0,2 s	0,2 s
Force	1,3 atm	1,3 atm	1,3 atm	1,3 atm	1,3 atm	1,3 atm
Power	750W	900W	1050W	1200W	1350W	600W



Ultrasonic welding of copper wires

Stan, Dorin[1a]; Cismaru, Mihai[1b]; Cosma, Anca-Daniela[2] Savu, Ionel-Dănuț[3]
stan.dorin.t2n@student.ucv.ro
[1a]Faculty of Mechanics, University of Craiova, RO, PhD student
cismaru.mihai.b2p@student.ucv.ro
[1b]Faculty of Mechanics, University of Craiova, RO, PhD student
cismaru.mihai@gmail.com
[2]University of Medicine and Pharmacy of Craiova, RO, PhD student
cosma.ancadanyela@yahoo.com
[3]Department of IMST, Faculty of Mechanics, University of Craiova, RO
ionel.savu@edu.ucv.ro

Abstract

Ultrasonic welding is one of the most efficient modern technologies for joining metallic materials, due to the fact that the process is carried out at temperatures much lower than the melting temperature of the base material. This characteristic leads to the reduction of heat-affected zones, the reduction of deformations and the preservation of the physical and mechanical properties of the joined components.

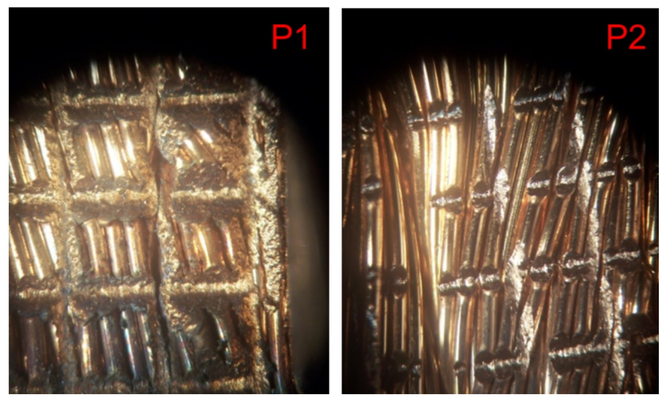
In the case of multi-wire electrical conductors, ultrasonic welding is of particular importance because it allows the creation of compact connections, with low electrical resistance and high mechanical stability. By compacting and consolidating the individual wires, uniform, reliable and vibration-resistant electrical contacts are obtained, which recommends the process for applications in the electrical, electronics and automotive industries, where the quality of electrical connections is essential for the safe operation of systems. However, the welding of copper conductors poses a number of technological difficulties due to the very high thermal and electrical conductivity of the material, which favours the rapid dissipation of the energy introduced into the joint area. Also, the formation of oxides on the surface of the conductors and the tendency of multi-strand wires to deform unevenly require rigorous control of the welding parameters to obtain stable and reproducible joints.

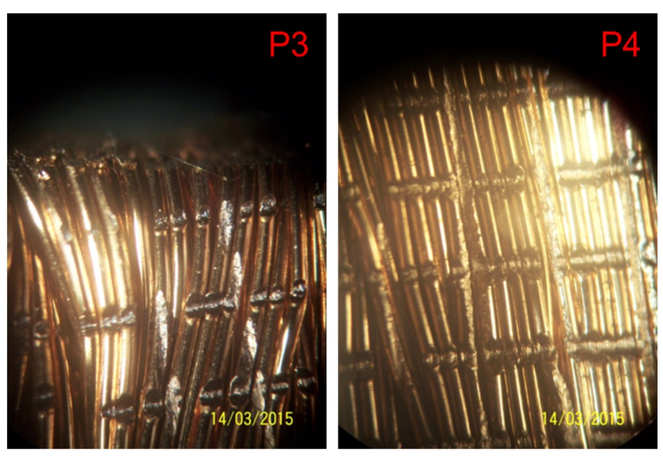
The paper investigates the influence of technological parameters on the quality of joints on multi-wire copper conductors, made by the ultrasonic welding process. The research investigated the theoretical behaviour of copper during welding and experimentally its behaviour during ultrasonic welding. T

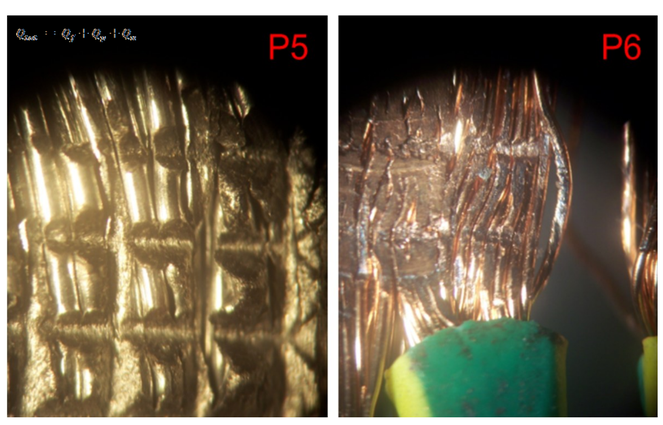
The experimental part highlighted the correlation between the amplitude of oscillations, the welding force, the activation time and the mechanical strength of the joints. The results obtained demonstrated that the use of welding regimes characterized by high ultrasonic powers and short times leads to the creation of stable connections, with minimal deformations and without significant structural damage to the base material. It was also found that the choice of the sonotrode material and the optimization of its geometry directly influence the efficiency of energy transfer and the durability of the equipment.

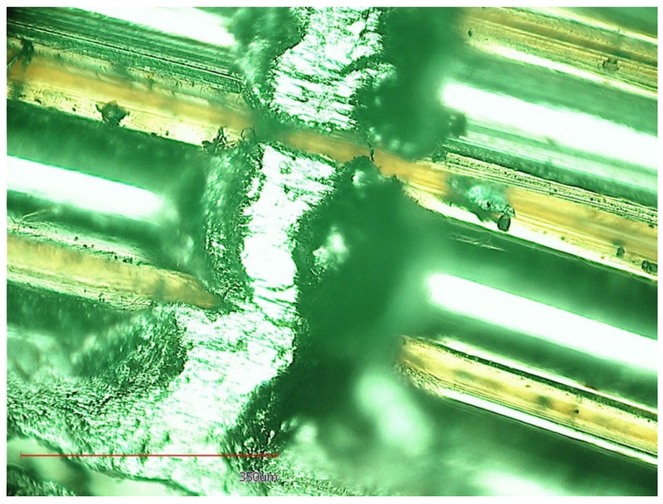
The research confirmed the applicability of ultrasonic technology for the creation of electrical connections used in the electrotechnical and automotive industries, where reliable, fast and easily automated joints are required. Experimental conclusions demonstrate that ultrasonic welding represents a high-performance alternative to conventional processes, offering important advantages in terms of productivity, energy consumption and the quality of the joints made.

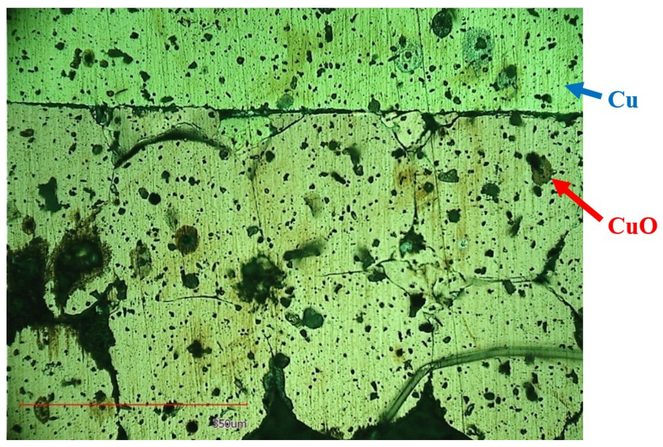
Results

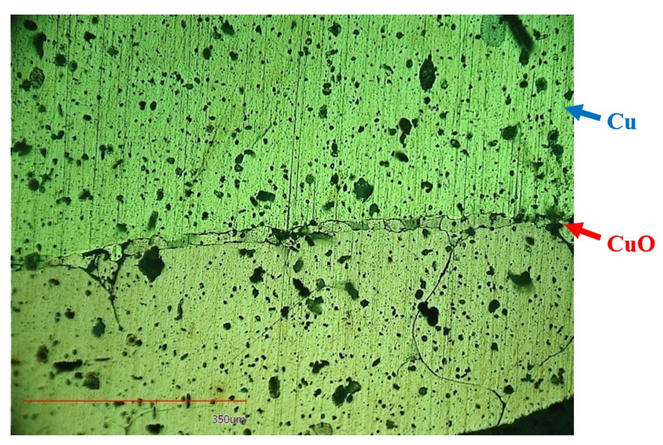


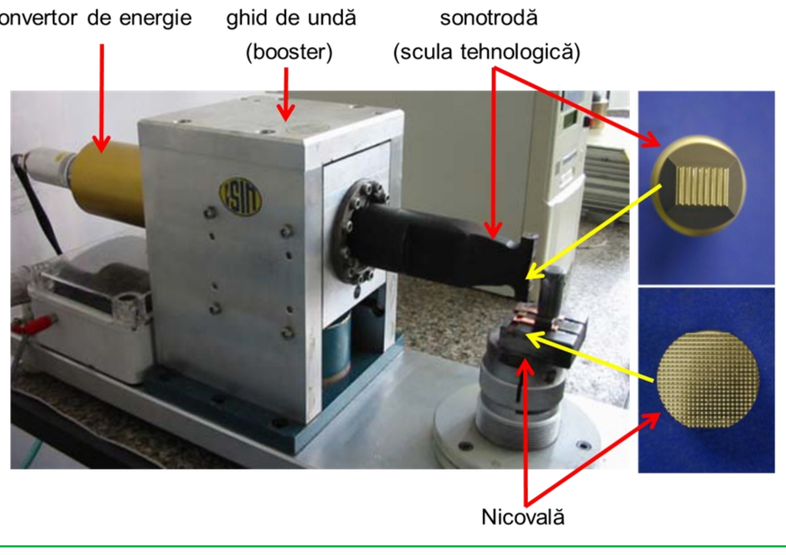


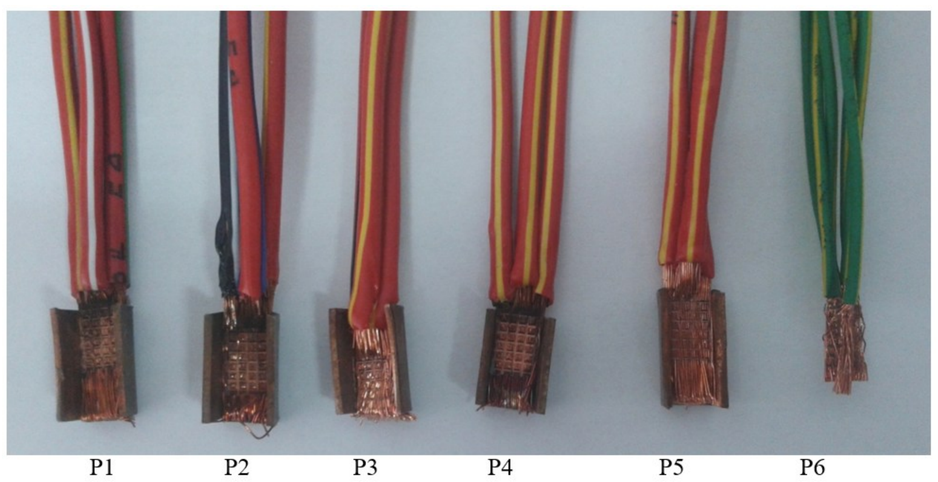












Welded specimens

Measured values for the electrical resistance

	P1	P2	P3	P4	P5	P6
Welded joints	161	156	155	146	145	-
Mechanical joints (5 measured values)			180 182 178 180 181	Average value: 180,2 ohm		
Diference	19,2	24,2	25,2	34,2	35,2	-